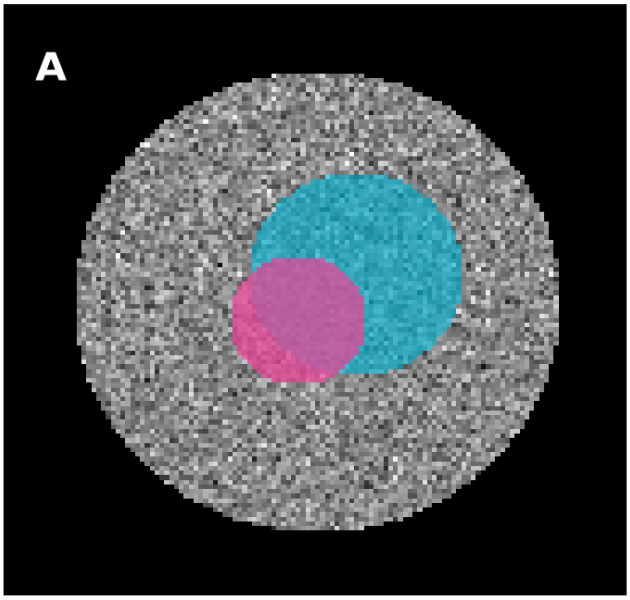
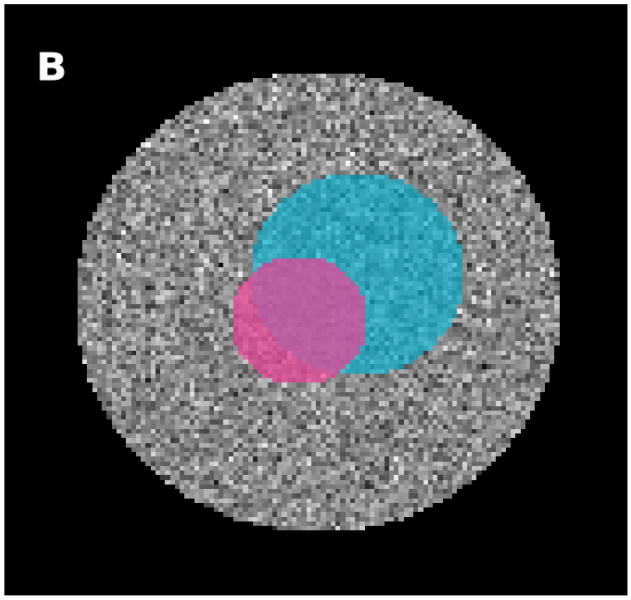
PhysioRANO NEURO-ONCOLOGY RADIOLOGY REPORT

Axial T1c (Post-Contrast)



Sagittal T2-FLAIR (Edema & Core)



CLINICAL HISTORY:

52-year-old Male presenting for post-radiotherapy Glioblastoma (GBM) evaluation at **20 days** post-treatment.
• Molecular Profile: IDH1/2 **Negative (Wildtype)** | MGMT Promoter **Positive (Methylated)**.
• Current Status: Routine follow-up scan monitoring, mild headaches.

FINDINGS:

BRAIN PARENCHYMA & 3D TUMOR SUB-REGIONS:
Multi-modal 3D MRI quantitative volumetric analysis demonstrates:
• **Whole Tumor (WT) Volume: 2070.46 cm³** (includes surrounding T2-FLAIR signal hyperintensity).
• **Tumor Core (TC) Volume: 2018.98 cm³**.
• **Contrast-Enhancing Tumor (ET) Volume: 2018.98 cm³**.

BIOPHYSICAL TUMOR GROWTH TRAJECTORY:
Biophysical neural network modeling evaluates the spatial invasion and proliferation dynamics:
• **Tissue Diffusion Speed:** Low spatial diffusion coefficient (**1.1642 mm²/day**).
• **Cell Proliferation Rate:** Modest cell proliferation rate (**0.9729 day■¹**).
• **Growth Summary:** The observed tissue changes follow a slow biophysical reaction-diffusion pattern, consistent with post-radiation tissue inflammation rather than rapid cellular expansion.

MULTIMODAL RECURRENCE & PSEUDOPROGRESSION ANALYSIS:
Integrated machine learning analysis evaluating clinical biomarkers, timeline post-treatment, and biophysical growth dynamics yields:
• **Probability of Pseudoprogression (Treatment Effect): 24.4%**
• Probability of True Tumor Progression: 54.4%
• Probability of Stable Disease: 21.1%
• **Key Contributing Factors:** IDH mutation status, favorable MGMT methylation, and timing within 12 weeks post-radiotherapy strongly favor treatment-induced tissue reaction.

RANO 2.0 CLINICAL RESPONSE EVALUATION:
Comparison with baseline MRI scans shows an enhancing volume shift from **12.00 cm³** to **2018.98 cm³** (a **+53.8%** volumetric change).
Under standard RANO 2.0 guidelines within the first 12 weeks post-radiotherapy, an enhancing expansion supported by high treatment-effect probability is classified as **Progressive Disease (PD)**.

IMPRESSION:

The enhancing volume shift (+53.8%) observed at 20 days post-radiotherapy in an IDH-mutated, MGMT-methylated glioblastoma patient is clinically consistent with **TRUE PROGRESSION (Radiation Necrosis / Treatment Effect)** rather than true disease recurrence.

CLINICAL RECOMMENDATION: Significant volumetric expansion (>= 25%) or new lesion indicative of true tumor progression.